

SYSTEMS ADMINISTRATION USING CISCO® (315)

STATE – 2019

TOTAL POINTS _____ *(500 points)*

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question.

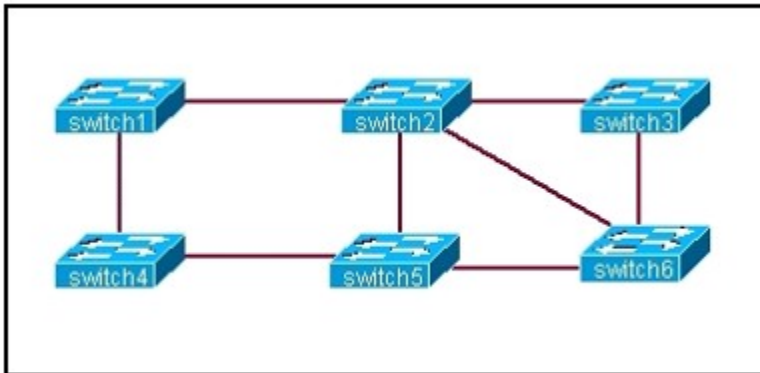
1. Which protocol advertises a virtual IP address to facilitate transparent failover of a Cisco routing device?
 - A. FHRP
 - B. DHCP
 - C. RSMLT
 - D. ESRP
2. Which protocol can cause overload on a CPU of a managed device?
 - A. Netflow
 - B. WCCP
 - C. IP SLA
 - D. SNMP
3. What protocol is the Cisco proprietary implementation of FHRP?
 - A. HSRP
 - B. VRRP
 - C. GLBP
 - D. CARP
4. Which parameter would you tune to affect the selection of a static route as a backup, when a dynamic protocol is also being used?
 - A. hop count
 - B. administrative distance
 - C. link bandwidth
 - D. link cost
5. What is the result of issuing the frame-relay map ip 192.168.1.2 202 broadcast command?
 - A. defines the destination IP address that is used in all broadcast packets on DLCI 202
 - B. defines the source IP address that is used in all broadcast packets on DLCI 202
 - C. defines the DLCI on which packets from 192.168.1.2 IP address are received
 - D. defines the DLCI that is used for all packets that are sent to the 192.168.1.2 IP address
6. What is a *valid* HSRP virtual MAC address?
 - A. 0000.5E00.01A3
 - B. 0007.B400.AE01
 - C. 0000.0C07.AC15
 - D. 0007.5E00.B301

7. Which command encrypts all plaintext passwords?
 - A. Router#service password-encryption
 - B. Router(config)#password-encryption
 - C. Router(config)#service password-encryption
 - D. Router#password-encryption
8. The command frame-relay map ip 10.121.16.8 102 broadcast was entered on the router. Which of the following statements is *true* concerning this command?
 - A. This command should be executed from the global configuration mode.
 - B. The IP address 10.121.16.8 is the local router port used to forward data.
 - C. This command is required for all Frame Relay configurations.
 - D. The broadcast option allows packets, such as RIP updates, to be forwarded across the PVC.
9. Three switches are connected to one another via trunk ports. Assuming the default switch configuration, which switch is elected as the root bridge for the spanning-tree instance of VLAN 1?
 - A. the switch with the highest MAC address
 - B. the switch with the lowest MAC address
 - C. the switch with the highest IP address
 - D. the switch with the lowest IP address
10. After you configure the Loopback0 interface, which command can you enter to verify the status of the interface and determine whether fast switching is enabled?
 - A. Router#show ip interface loopback 0
 - B. Router#show run
 - C. Router#show interface loopback 0
 - D. Router#show ip interface brief
11. When a router undergoes the exchange protocol within OSPF, in what order does it pass through each state?
 - A. exstart state > loading state > exchange state > full state
 - B. exstart state > exchange state > loading state > full state
 - C. exstart state > full state > loading state > exchange state
 - D. loading state > exchange state > full state > exstart state
12. What command is used to verify the DLCI destination address in a Frame Relay static configuration?
 - A. show frame-relay pvc
 - B. show frame-relay lmi
 - C. show frame-relay map
 - D. show frame relay end-to-end

13. A network engineer wants to allow a temporary entry for a remote user with a specific username and password so that the user can access the entire network over the Internet. Which ACL can be used?
- A. standard
 - B. extended
 - C. dynamic
 - D. reflexive
14. Which command can you enter to configure a local username with an encrypted password and EXEC mode user privileges?
- A. Router(config)#username jdone privilege 1 password 7 08314D5D1A48
 - B. Router(config)#username jdone privilege 1 password 7 PASSWORD1
 - C. Router(config)#username jdone privilege 15 password 0 08314D5D1A48
 - D. Router(config)#username jdone privilege 15 password 0 PASSWORD1
15. What will be the result if the following configuration commands are implemented on a Cisco switch?
- Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security mac-address sticky
- A. A dynamically learned MAC address is saved in the startup-configuration file.
 - B. A dynamically learned MAC address is saved in the running-configuration file.
 - C. A dynamically learned MAC address is saved in the VLAN database.
 - D. Statically configured MAC addresses are saved in the startup-configuration file if frames from that address are received.
16. What is known as “one-to-nearest” addressing in IPv6?
- A. global unicast
 - B. anycast
 - C. multicast
 - D. unspecified address
17. The network administrator needs to address seven LANs. RIP version 1 is the only routing protocol in use on the network and subnet 0 is not being used. What is the maximum number of usable IP addresses that can be supported on each LAN if the organization is using one class C address block?
- A. 8
 - B. 6
 - C. 30
 - D. 32

18. Cisco Catalyst switches CAT1 and CAT2 have a connection between them using ports FA0/13. An 802.1Q trunk is configured between the two switches. On CAT1, VLAN 10 is chosen as native, but on CAT2 the native VLAN is not specified. What will happen in this scenario?
- A. 802.1Q giants frames could saturate the link.
 - B. VLAN 10 on CAT1 and VLAN 1 on CAT2 will send untagged frames.
 - C. A native VLAN mismatch error message will appear.
 - D. VLAN 10 on CAT1 and VLAN 1 on CAT2 will send tagged frames.
19. A network administrator creates a layer 3 EtherChannel, bundling four interfaces into channel group 1. On what interface is the IP address configured?
- A. the port-channel 1 interface
 - B. the highest number member interface
 - C. all member interfaces
 - D. the lowest number member interface
20. Which condition indicates that service password-encryption is enabled?
- A. The local username password is encrypted in the configuration.
 - B. The enable secret is encrypted in the configuration.
 - C. The local username password is in clear text in the configuration.
 - D. The enable secret is in clear text in the configuration.
21. What command instructs the device to timestamp Syslog debug messages in milliseconds?
- A. service timestamps log datetime localtime<input type
 - B. service timestamps debug datetime msec<input type
 - C. service timestamps debug datetime localtime<input type
 - D. service timestamps log datetime msec
22. VLAN 3 is *not* yet configured on your switch. What happens if you set the switchport access vlan 3 command in interface configuration mode?
- A. The command is rejected.
 - B. The port turns amber.
 - C. The command is accepted and the respective VLAN is added to vlan.dat.
 - D. The command is accepted and you must configure the VLAN manually.
23. Which Cisco Catalyst feature automatically disables the port in an operational PortFast upon receipt of a BPDU?
- A. BackboneFast
 - B. UplinkFast
 - C. Root Guard
 - D. BPDU Guard

24. Which set of command is recommended to prevent the use of a hub in the access layer?
- A. switch(config-if)#switchport mode trunk
switch(config-if)#switchport port-security maximum 1
 - B. switch(config-if)#switchport mode trunk
switch(config-if)#switchport port-security mac-address 1
 - C. switch(config-if)#switchport mode access
switch(config-if)#switchport port-security maximum 1
 - D. switch(config-if)#switchport mode access
switch(config-if)#switchport port-security mac-address 1
25. Which PPP subprotocol negotiates authentication options?
- A. NCP
 - B. ISDN
 - C. SLIP
 - D. LCP
26. If primary and secondary root switches with priority 16384 both experience catastrophic losses, which tertiary switch can take over?
- A. a switch with priority 20480
 - B. a switch with priority 8192
 - C. a switch with priority 4096
 - D. a switch with priority 12288
27. Based on the network shown in the graphic. Which option contains both the potential networking problem and the protocol or setting that should be used to prevent the problem?



- A. routing loops, hold down timers
- B. switching loops, STP
- C. switching loops, VTP
- D. routing loops, split horizon

28. What SNMP message alerts the manager to a condition on the network?
- A. response
 - B. get
 - C. trap
 - D. capture
29. What command is used to display the collection of OSPF link states?
- A. show ip ospf link-state
 - B. show ip ospf lsa database
 - C. show ip ospf neighbors
 - D. show ip ospf database
30. Which command can you enter to view the ports that are assigned to VLAN 20?
- A. Switch#show vlan id 20
 - B. Switch#show ip interface brief
 - C. Switch#show interface vlan 20
 - D. Switch#show ip interface vlan 20
31. What is the effect of the overload keyword in a static NAT translation configuration?
- A. It enables port address translation.
 - B. It enables the use of a secondary pool of IP addresses when the first pool is depleted.
 - C. It enables the inside interface to receive traffic.
 - D. It enables the outside interface to forward traffic.
32. Which term describes a spanning-tree network that has all switch ports in either the blocking or forwarding state?
- A. converged
 - B. redundant
 - C. provisioned
 - D. spanned
33. Which parameter or parameters are used to calculate OSPF cost in Cisco routers?
- A. Bandwidth
 - B. Bandwidth and Delay
 - C. Bandwidth, Delay, and MTU
 - D. Bandwidth, MTU, Reliability, Delay, and Load
34. Assuming the default switch configuration, which VLAN range can be added, modified, and removed on a Cisco switch?
- A. 1 through 1001
 - B. 2 through 1001
 - C. 1 through 1002
 - D. 2 through 1005

35. Which standards-based First Hop Redundancy Protocol is a Cisco supported alternative to Hot Standby Router Protocol?
- A. GLBP
 - B. TFTP
 - C. VRRP
 - D. DHCP
36. Which switch would STP choose to become the root bridge in the selection process?
- A. 32768: 22-33-44-55-66-77
 - B. 32769: 11-22-33-44-55-65
 - C. 32769: 22-33-44-55-66-78
 - D. 32768: 11-22-33-44-55-66
37. Which statement about RADIUS security is *true*?
- A. It supports EAP authentication for connecting to wireless networks.
 - B. It provides encrypted multiprotocol support.
 - C. Device-administration packets are encrypted in their entirety.
 - D. It ensures that user activity is fully anonymous.
38. Which command can you enter to determine whether serial interface 0/2/0 has been configured using HDLC encapsulation?
- A. router#show platform
 - B. router#show interfaces Serial 0/2/0
 - C. router#show ip interface s0/2/0
 - D. router#\show ip interface brief
39. What authentication type is used by SNMPv2?
- A. HMAC-MD5
 - B. HMAC-SHA
 - C. CBC-DES
 - D. community strings
40. What is the default Syslog facility level?
- A. local4
 - B. local5
 - C. local6
 - D. local7
41. Which protocol is an open standard protocol framework that is commonly used in VPNs, to provide secure end-to-end communications?
- A. RSA
 - B. L2TP
 - C. IPsec
 - D. PPTP

42. During which phase of PPPoE is PPP authentication performed?
- A. the PPP Session phase
 - B. Phase 2
 - C. the Active Discovery phase
 - D. the Authentication phase
43. Which Layer 2 protocol encapsulation type supports synchronous and asynchronous circuits and has built-in security mechanisms?
- A. HDLC
 - B. PPP
 - C. X.25
 - D. Frame Relay
44. When an interface is configured with PortFast BPDU guard, how does the interface respond when it receives a BPDU?
- A. It continues operating normally.
 - B. It goes into a down/down state.
 - C. It becomes the root bridge for the configured VLAN.
 - D. It goes into an Errdisable State.
45. Why will a switch never learn a broadcast address?
- A. Broadcasts only use network layer addressing.
 - B. A broadcast frame is never forwarded by a switch.
 - C. A broadcast address will never be the source address of a frame.
 - D. Broadcast frames are never sent to switches.
46. Which command is used to enable CHAP authentication, with PAP as the fallback method, on a serial interface?
- A. Router(config-if)#ppp authentication chap fallback ppp
 - B. Router(config-if)#ppp authentication chap pap
 - C. Router(config-if)#authentication ppp chap fallback ppp
 - D. Router(config-if)#authentication ppp chap pap
47. Which command can you enter to verify that a BGP connection to a remote device is established?
- A. show ip bgp summary
 - B. show ip community-list
 - C. show ip bgp paths
 - D. show ip route
48. Which feature builds a FIB and an adjacency table to expedite packet forwarding?
- A. Cisco Express Forwarding
 - B. process switching
 - C. fast switching
 - D. cut-through

49. Which command is necessary to permit SSH or Telnet access to a Cisco switch that is otherwise configured for these vty line protocols?
- A. transport type all
 - B. transport output all
 - C. transport preferred all
 - D. transport input all
50. In which circumstance are multiple copies of the same unicast frame likely to be transmitted in a switched LAN?
- A. during high traffic periods
 - B. after broken links are re-established
 - C. when upper-layer protocols require high reliability
 - D. in an improperly implemented redundant topology